Arun Vignesh Malarkkan

malarkkanarunvignesh@gmail.com (+1) 480-255-2954 http://arunvignesh28.github.io/

About Me

- Researching the reliability and generalization of LLM reasoning using causal inference and mechanistic interpretability, focusing on invariance discovery, spurious correlation mitigation, and robust evaluation.

Education

Aug 2023 – Present
- **Ph.D., Computer Science**, Arizona State University, Tempe, AZ.
  Advisor: Dr. Yanjie Fu. Knowledge Discovery and Data Mining Lab.
  CGPA: 3.9/4.0.

May 2020
- **Master of Science, Computer Science**, Arizona State University, Tempe, AZ.
  CGPA: 3.8/4.0.

Technical Skills

Core Expertise	LLM Reasoning & Reliability, Causal Inference & Causal Representation Learning, Mechanistic Interpretability, Transformer Architectures, Post-training & Fine-tuning, Retrieval-Augmented Generation (RAG), Data-Centric AI.
ML & Systems	PyTorch, TensorFlow, Hugging Face Transformers, DeepSpeed, Distributed Training, GPU/CUDA Programming.
LLM & Generative AI	PEFT (LoRA, QLoRA), LangChain, LlamaIndex, OpenAI API, vLLM, Vertex AI, Weights & Biases (W&B).
Interpretability & Analysis	TransformerLens, Activation Patching, Probing (linear probes, representation analysis), PyTorch Hooks.
Programming	Python, Java, TypeScript.
Data & Infrastructure	AWS (SageMaker, RDS, DynamoDB), MySQL, Apache Spark (SparkML), Docker.
General Libraries	Scikit-learn, Pandas, NumPy, Flask, AutoGluon.

Research Publications

First-Author Publications

1. **A. V. Malarkkan**, W. Ying, and Y. Fu, *CAFE: Causally-guided automated feature engineering with multi-agent reinforcement learning*, arXiv preprint. Under review, ICML, 2026.
2. **A. V. Malarkkan** et al., *FinRule-Bench: A benchmark for joint reasoning of llms over financial tables and principles*, arXiv preprint. Under review, KDD, 2026.
3. **A. V. Malarkkan**, H. Bai, A. Kaushik, and Y. Fu, "DELTA: Variational disentangled learning for privacy-preserving data reprogramming," in *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 2025.
4. **A. V. Malarkkan**, H. Bai, D. Wang, and Y. Fu, "Causal graph profiling via structural divergence for robust anomaly detection in cyber-physical systems," in *Proceedings of the IEEE International Conference on Big Data*, 2025.
5. **A. V. Malarkkan**, D. Wang, H. Bai, and Y. Fu, "Incremental causal graph learning for online cyberattack detection in cyber-physical infrastructures," *IEEE Transactions on Big Data*, 2025.
6. **A. V. Malarkkan**, D. Wang, and Y. Fu, "Multi-view causal graph fusion based anomaly detection in cyber-physical infrastructures," in *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*, 2024.

Co-Authored Publications

1. H. Cao, X. Wang, **A. V. Malarkkan**, K. Liu, D. Wang, and Y. Fu, *Rethinking data augmentation under covariate shift: Invariant-guided diffusion and prototype reweighting*, Under review. Under review, KDD, 2026.
2. X. Wang, K. Liu, **A. V. Malarkkan**, and Y. Fu, *Evolving demonstration optimization for chain-of-thought feature transformation*, arXiv preprint. Under review, KDD, 2026.
3. W. Ying et al., *Distribution shift aware neural tabular learning*, arXiv preprint. Under review, ICML, 2026.
4. W. Ying et al., *A survey on data-centric ai: Tabular learning from reinforcement learning and generative ai*, arXiv preprint., 2025.

Professional Experience

Jun 2025 – Present

Doctoral Researcher, DOW Chemicals (National Academy of Engineering Grant).

- Leading research on incorporating LLM-guided knowledge priors into neuro-symbolic, causally-aware multi-agent reinforcement learning frameworks to encode domain constraints and guide structured reasoning in materials discovery.
- Developing frameworks that integrate causal structure with sequential decision-making to improve robustness under distribution shifts in industrial simulation environments.
- Designing interpretable and scalable models for material optimization under uncertainty (details subjected to confidentiality).

Jan 2023 – Jun 2023

Senior Software Engineer, Fidelity Investments.

- Led development of Monte Carlo-based financial projection systems for retirement planning at scale.
- Built probabilistic simulation frameworks for uncertainty-aware decision modeling in long-term financial forecasting.
- Developed ensemble-based outlier detection models to identify key drivers affecting projected savings outcomes.

Jun 2021 – Jan 2023

Software Development Engineer, Amazon — Alexa AI.

- Developed and deployed purchase likelihood modeling for Alexa recommendation systems, improving voice-driven purchases by $+8\%$.
- Built end-to-end decision and ranking pipelines for voice-based commerce, integrating user behavior modeling with real-time inference systems.
- Designed production systems for voice-enabled transactional flows with compliance constraints (e.g., parental consent, HIPAA-aligned systems).

Jun 2020 – May 2021

Data Scientist / Software Developer, ASU Decision Center for Educational Excellence.

- Developed predictive models for demographic, financial, and infrastructure planning using large-scale public datasets.
- Designed end-to-end data pipelines (AWS, ETL) for data ingestion, modeling, and deployment.
- Built models for resource allocation and policy analysis, identifying key drivers of graduation rates and community-level outcomes.

Academic Research Experience

2024 – Present

LLM Reasoning: Causal Mediation and Invariant Mechanisms

- Investigating whether LLMs encode causally invariant reasoning mechanisms or rely on spurious correlations, using interventional analysis over Transformer computations.
- Developed a causal mediation framework over intermediate tokens, combining activation patching and counterfactual inputs to estimate mediated effects on model predictions, demonstrating that only a sparse subset of tokens act as stable causal mediators.
- Identified a mechanistic view of reasoning as a multi-stage process: candidate formation, causal mediation, redundancy, and invariant consolidation.

2025 – 2026

FinRule-Bench: Benchmarking LLM Reasoning under Rule Constraints (Under review — KDD 2026)

- Developed a benchmark for evaluating rule-based reasoning and diagnostic completeness in LLMs over structured financial data.
- Introduced causal-counterfactual evaluation protocols to assess reasoning consistency and failure modes.

2024 – 2026

Causally-Guided Automated Feature Engineering with Multi-Agent Reinforcement Learning (Under review — ICML 2026)

- Developed a causal-RL framework for feature construction using causal graphs as inductive priors.
- Designed structured exploration and reward shaping for robustness under distribution shift.

2024 – 2025

Privacy-Preserving Generative Data Reprogramming (Accepted — ICDM 2025)

- Proposed a variational generative framework with causal disentanglement for privacy-aware feature transformation.
- Achieved ∼9% performance improvement and ∼35% reduction in privacy leakage.

2023 – 2025

Research on Causal Learning and Robustness

- Conducted research on causal representation learning and invariant modeling for improving robustness under distribution shift, focusing on disentangling stable mechanisms from spurious correlations across learning systems.
- Developed incremental and multi-view causal graph learning frameworks for robust modeling under distribution shifts and streaming environments (IEEE TBD, IEEE Big Data, CIKM).
- Focused on invariant structure learning and continual adaptation in non-stationary systems.

Teaching, Mentorship & Academic Service

Invited Talks

Dec 2025

Keynote — IEEE International Conference on Modelling, Simulation & Intelligent Computing (MoSICom), BITS Pilani, UAE.
"Beyond Correlation: Can Causal Reasoning Enhance Modern AI Models in Robustness and Interpretability?"
Presented research on causal inference, robustness, and interpretability in modern AI systems, emphasizing limitations of correlation-based learning and the role of causal reasoning.

Teaching, Mentorship & Academic Service (continued)

Teaching

Aug 2023 – Present

Senior Teaching Assistant / Lab Instructor, Arizona State University.

- Led instruction and lab sessions for courses in Object-Oriented Programming, Data Mining, and Semantic Web Mining for large undergraduate cohorts, and also project management for 30 Capstone Project teams.
- Designed and developed hands-on lab modules and course materials, emphasizing applied machine learning and data-driven reasoning.

Mentorship

Jan 2024 – Present

Research Mentor, Arizona State University.

- Mentoring undergraduate and graduate researchers on Neuro-symbolic AI for feature engineering, world model simulations, and LLM reasoning optimization.
- Guiding projects on investigating efficient fine-tuning and evaluation of large language models for reasoning tasks, including experimental design and model analysis.

Academic Service

Jan 2025 – Present

Program Committee Member: AAAI 2026; IEEE ICSCAN 2024; IEEE BigData 2024 (Undergraduate & High School Research Symposium); IEEE ICDM 2025.
Conference Reviewer: KDD, ICLR, NeurIPS, ICDM, IEEE BigData, Urban AI.
Journal Reviewer: ACM Transactions on Knowledge Discovery from Data (TKDD).
Session Chair: IEEE MoSICom 2025; IEEE ICDM 2025 — Undergraduate & High School Research Symposium.

Workshops & Outreach

Outreach

- Conducted hands-on workshops on Machine Learning fundamentals and Graph Neural Networks, training undergraduate students in applied AI techniques.
- Developed instructional material focused on practical model development and real-world applications.
- IEEE Phoenix Chapter Student Volunteer & Organizer

Honors & Awards

2025 **IEEE ICDM Scholarship**, IEEE International Conference on Data Mining, 2025.

2018 **Academic Scholarship**, Arizona State University, 2018.

2017–2018 **Best Undergraduate Researcher Award**, 2017–2018.